Bank Customer Churn Analysis

Project Overview

This project analyses customer churn in a banking dataset to understand customer retention patterns, identify customer segments with relatively high churn rates, and generate actionable business insights.

Customer churn occurs when customers discontinue their relationship with a business. For banks, understanding churn is important because retaining existing customers can support long-term revenue, customer relationships, and business stability.

The project uses Python for data cleaning and exploratory data analysis (EDA), and Microsoft Power BI for interactive data visualization and dashboard development.

The final deliverable is a two-page interactive dashboard that presents customer churn patterns across demographic, geographic, and behavioural characteristics.

Project Objectives

The main objectives were to:

- Examine the bank's customer base and overall churn rate.
- Identify geographic and demographic segments with higher churn rates.
- Investigate the relationship between customer activity and churn.
- Examine churn patterns across different numbers of bank products.
- Compare customer balances between retained and churned customers.
- Develop an interactive Power BI dashboard to communicate findings.
- Provide practical recommendations that could support customer retention strategies.

Tools and Technologies

Tool	Purpose
Python	Data preparation and analysis
Pandas	Data cleaning, manipulation, and aggregation
NumPy	Numerical operations
Jupyter Notebook	Interactive analysis and documentation of the workflow
Microsoft Power BI	Interactive dashboard development
DAX	Measures and calculated columns
Excel / CSV	Source data inspection and data storage

Dataset Description

The dataset consisted of two Excel worksheets:

1. **Customer Info:** Customer demographic and personal information.
2. **Account Info:** Customer account and banking information.

The two datasets were cleaned and combined using the customer identifier.

The final dataset contained **10,000 customer records and 13 columns**.

Important variables

Variable	Description
CustomerId	Unique customer identifier
Surname	Customer surname
CreditScore	Customer credit score
Geography	Customer's country
Gender	Customer gender
Age	Customer age
Tenure	Number of years with the bank
Balance	Customer account balance
NumOfProducts	Number of bank products held
HasCrCard	Whether the customer has a credit card
IsActiveMember	Whether the customer is an active member
EstimatedSalary	Estimated customer salary
Exited	Whether the customer left the bank (1 = Yes, 0 = No)

Data Cleaning and Preparation

Data cleaning was performed using Python and Pandas before the dataset was imported into Power BI.

The major preparation steps included:

1. Data inspection

- Loaded both worksheets.
- Examined dataset dimensions, column names, data types, and sample records.
- Identified missing values, duplicate records, inconsistent formats, and invalid entries.

2. Duplicate removal

Duplicate records and duplicate customer identifiers were investigated and removed to avoid counting customers more than once.

3. Missing-value treatment

- Missing surnames were replaced with `Unknown`.
- Missing Age values were imputed using the median.
- Missing EstimatedSalary values were handled after correcting invalid placeholder entries.

4. Salary cleaning

The EstimatedSalary column contained currency symbols, commas, and invalid placeholder values.

These were cleaned by:

- Removing currency symbols and commas.
- Converting the column to a numeric data type.
- Replacing the `-999999` placeholder with a missing value.
- Imputing missing values using the median.

5. Standardization and data types

- Standardized Geography values into France, Germany, and Spain.
- Converted Balance to a numeric data type.
- Converted HasCrCard and IsActiveMember from Yes/No values into binary indicators (1/0).
- Checked Tenure across the worksheets and retained a single appropriate column.
- Verified the Exited variable as a binary churn indicator.

6. Dataset integration

The cleaned Customer Info and Account Info datasets were merged using CustomerId.

The resulting dataset contained 10,000 records and 13 columns and was exported as `bank_churn_cleaned.csv` for use in Power BI.

Exploratory Data Analysis (EDA)

EDA was conducted in Python through four major areas: univariate, bivariate, multivariate, and correlation analysis.

1. Univariate analysis

This stage examined individual variables to understand the composition of the customer base.

Key observations:

- France accounted for 50.14% of customers.
- Germany accounted for 25.09%.
- Spain accounted for 24.77%.
- Male customers represented 54.57% of the dataset, while female customers represented 45.43%.
- Approximately 70.55% of customers had a credit card.
- Approximately 51.51% were active members.
- Most customers held either one or two bank products.

Overall churn distribution:

Customer status	Customers	Percentage
Retained	7,963	79.63%
Churned	2,037	20.37%
Total	10,000	100%

The overall churn rate was **20.37%**.

2. Bivariate analysis

This stage examined relationships between individual customer characteristics and churn.

Geography and churn

Geography	Customers churned	Churn rate
France	810	16.15%
Germany	814	32.44%
Spain	413	16.67%

Germany had the highest churn rate, approximately twice the rate observed in France and Spain.

Gender and churn

Gender	Churn rate
Female	25.07%
Male	16.46%

Female customers had a higher observed churn rate than male customers.

Age group and churn

Age group	Churn rate
18–25	7.53%
26–35	8.50%
36–45	19.62%
46–55	50.57%
56–65	48.32%
66+	13.26%

Customers aged 46–55 and 56–65 recorded the highest churn rates among the age groups examined.

Active membership and churn

Membership status	Churn rate
Active	14.27%
Inactive	26.85%

Inactive members had a substantially higher churn rate than active members.

Number of products and churn

Number of products	Churn rate
1	27.71%
2	7.58%
3	82.71%
4	100%

Customers with three or four products showed unusually high churn rates. However, these groups were small: 266 customers held three products and only 60 held four. These percentages should therefore be interpreted with caution.

Other numerical variables

Churned customers had:

- A higher average age (44.84 years) than retained customers (37.41 years).
- A higher average account balance (91,108.54) than retained customers (72,745.30).
- A slightly lower average credit score (645.35 versus 651.85).
- Very similar average tenure (4.93 versus 5.03 years).
- A slightly higher average estimated salary (101,465.68 versus 99,738.39).

Credit-card ownership showed little difference in churn rates: 20.18% for customers with a card and 20.81% for those without one.

3. Multivariate analysis

Multivariate analysis examined combinations of customer characteristics to identify more detailed patterns.

Important observations included:

- Germany had higher churn rates across both male and female customer segments.
- Female customers had higher churn rates than male customers across all three geographic regions.
- Inactive customers generally had higher churn rates than active customers across the geographic and age segments examined.
- Customers aged 46–55 and 56–65 showed particularly high churn rates, especially among inactive members.
- Customers holding three products showed high churn rates across all three countries.

For example, among customers aged 56–65, the observed churn rates for inactive members were particularly high in France, Germany, and Spain. Some of these subgroup results are based on relatively small customer counts and should be interpreted accordingly.

4. Correlation analysis

Correlation analysis was used to examine linear relationships between numerical variables and the Exited variable.

The most notable correlations with churn were:

Variable	Correlation with Exited
Age	+0.285
Balance	+0.119
IsActiveMember	-0.156
CreditScore	-0.027
NumOfProducts	-0.048
EstimatedSalary	+0.012
Tenure	-0.014
HasCrCard	-0.007

Age had the strongest positive linear correlation with churn among the listed customer characteristics. Active membership had a negative correlation, indicating that active customers were less likely to have exited in this dataset.

Correlation values indicate association, not causation. A low linear correlation does not necessarily mean a variable has no useful relationship with churn. For example, the number-of-products analysis revealed substantial differences between product categories that a single correlation coefficient does not adequately describe.

Power BI Dashboard

An interactive two-page dashboard was developed to communicate the analysis.

Page 1: Customer Churn Overview

This page provides a high-level summary of customer retention and churn.

It includes:

- Total Customers
- Churned Customers
- Retained Customers
- Churn Rate
- Retention Rate
- Churned Customers by Geography
- Churn Rate by Geography
- Churn Rate by Gender
- Churn Rate by Age Group
- Churn Rate by Active Membership Status

Interactive slicers allow users to explore customer segments by geography, gender, age group, and membership status.

Page 2: Customer Behaviour and Churn Drivers

This page explores customer characteristics and their observed relationship with churn.

It includes:

- Churn Rate by Number of Products
- Average Customer Balance by Churn Status
- Churned Customers by Geography and Gender
- Customer Distribution by Geography
- Churn Rate by Credit Card Ownership

These visuals provide additional context for interpreting the patterns identified during EDA.

Key DAX Measures

The following measures were created in Power BI:

Total Customers

```
Total Customers =
COUNTROWS('bank_churn_cleaned')
```

Churned Customers

```
Churned Customers =
CALCULATE(
    COUNTROWS('bank_churn_cleaned'),
    'bank_churn_cleaned'[Exited] = 1
)
```

Retained Customers

```
Retained Customers =
CALCULATE(
    COUNTROWS('bank_churn_cleaned'),
    'bank_churn_cleaned'[Exited] = 0
)
```

Churn Rate

```
Churn Rate =
DIVIDE(
    [Churned Customers],
    [Total Customers],
    0
)
```

Retention Rate

```
Retention Rate =
DIVIDE(
    [Retained Customers],
    [Total Customers],
    0
)
```

Average Balance

```
Average Balance =
AVERAGE('bank_churn_cleaned'[Balance])
```

An Age Group calculated column was also created to support age-based analysis in Power BI.

Key Findings

The analysis produced several important findings:

1. **Overall churn:** 2,037 out of 10,000 customers had exited, representing a churn rate of 20.37%.
2. **Geographic differences:** Germany recorded the highest churn rate at 32.44%, compared with 16.15% in France and 16.67% in Spain.
3. **Age-related patterns:** Customers aged 46–55 and 56–65 recorded the highest churn rates, at 50.57% and 48.32%, respectively.
4. **Gender differences:** Female customers had a higher observed churn rate (25.07%) than male customers (16.46%).
5. **Customer engagement:** Inactive members had a churn rate of 26.85%, compared with 14.27% among active members.

6. **Product ownership:** Customers with three and four products showed exceptionally high churn rates, although these groups represented a small portion of the dataset.
7. **Account balance:** Churned customers had a higher average balance than retained customers.
8. **Credit-card ownership:** Credit-card ownership showed almost no meaningful difference in churn rates.

Business Insights

The findings suggest several areas where the bank could investigate customer retention opportunities.

Geographic retention opportunities

Germany should receive particular attention because it recorded the highest churn rate. The bank could investigate whether customer experience, service quality, pricing, product suitability, or regional market conditions differ from those in France and Spain.

Customer engagement

The higher churn rate among inactive members suggests that customer engagement may be an important area to investigate. Reduced account activity could be a useful indicator for identifying customers who may need additional attention.

Age-related customer retention

The elevated churn rates among customers aged 46–65 suggest that the bank should investigate whether its products and services meet the expectations of these customer segments.

Product experience

The unusually high churn rates among customers with three or four products deserve further investigation. The bank should examine whether these customers experience product complexity, service problems, unsuitable product combinations, or other issues.

Customer balance

The higher average balance among churned customers suggests that customer retention efforts should not focus only on the number of customers leaving. The bank should also assess the potential financial importance of customers who leave.

Recommendations

Based on the descriptive findings, the following actions are recommended:

1. **Investigate churn in Germany:** Conduct a deeper review of customer feedback, service experience, product usage, and regional banking conditions.
2. **Develop customer engagement programmes:** Identify inactive customers and consider targeted communication, personalized services, and relevant engagement campaigns.
3. **Prioritize high-churn age segments:** Investigate the needs and expectations of customers aged 46–65 and develop retention initiatives based on further customer research.
4. **Review multi-product customer experiences:** Examine the reasons for high churn among customers with three or four products before making changes to product bundles or sales strategies.

5. **Monitor customer account activity:** Track changes in customer activity and establish an early-warning process for customers showing signs of disengagement.
6. **Investigate high-balance customer exits:** Review the circumstances surrounding the departure of customers with higher account balances and assess the potential financial impact.
7. **Build a churn monitoring process:** Regularly monitor churn rate, retention rate, customer activity, and geographic performance through an updated dashboard.

These recommendations should be tested through further investigation and measured retention initiatives before being treated as proven solutions.

Limitations

This project is a descriptive analysis of the supplied customer dataset. Several limitations should be considered:

- The analysis identifies associations and patterns but does not establish causation.
- Some customer segments, particularly those holding three or four products, have relatively small sample sizes.
- Median imputation was used for selected missing numerical values, which may affect the distribution of those variables.
- The dataset does not provide all possible explanations for customer departures, such as customer complaints, competitor offers, service interactions, or detailed reasons for account closure.
- Correlation analysis captures linear relationships and may not fully represent more complex patterns.
- The analysis does not yet include statistical significance testing or predictive modelling.

Further work could include statistical hypothesis testing, customer segmentation, and the development of a machine-learning churn prediction model.

Conclusion

This project demonstrated how Python and Power BI can be used together to transform raw banking data into meaningful analytical insights.

After cleaning and integrating two datasets, 10,000 customer records were analysed to examine churn patterns across geography, age, gender, account balance, customer activity, and product ownership.

The analysis identified an overall churn rate of 20.37%, with Germany, customers aged 46–65, female customers, inactive members, and certain multi-product customer groups showing relatively high churn rates.

The Power BI dashboard provides an interactive way to explore these findings and supports a more informed approach to customer retention analysis.

Overall, the project strengthened practical skills in data cleaning, exploratory data analysis, descriptive statistics, DAX, data visualization, and business insight communication. It also provides a foundation for future work involving statistical analysis and machine-learning-based customer churn prediction.

Project Type: Data Analytics

Tools: Python, Pandas, NumPy, Jupyter Notebook, Microsoft Power BI, DAX

Final Dataset: 10,000 records × 13 columns

Dashboard: Two interactive report pages